VaaniQ: Cross-Lingual, Compression-Robust Detection and Calibrated Reliability Estimation for AI-Generated Voice in Indian Languages

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IEEE archival draft — frozen Round 3 evidence at commit 084bd47ca6ca1b69a7cdbf424e2946f3794c2a95. RQ1–RQ4 complete on the bounded machine-evaluated benchmark.

Abstract—Voice cloning enables scalable audio fraud, yet anti-spoofing evidence remains concentrated in English/Mandarin and rarely couples Indian-language stimuli with messaging-style compression, confidence calibration, and cross-lingual transfer. VaaniQ builds a bounded speaker-disjoint benchmark for Hindi, Marathi, and Tamil (Kathbath real; IndicSynth fake) and evaluates detection under clean audio and WhatsApp-style 16 kbps Opus simulation. On 584 held-out instances, an acoustic AASIST-compatible head reaches 91.61% accuracy and 6.56% EER; frozen wav2vec2-xls-r-300m reaches 92.12% accuracy, 6.88% EER, and 0.9828 ROC-AUC. Compression effects are model-dependent; English-only transfer collapses; leave-one-language-out transfer is asymmetric; validation-selected calibration does not uniformly improve held-out ECE. Benchmark V2 remains a partial external-source pilot, so claims are restricted to the declared machine-evaluated corpus and attack sources.

Index Terms—audio deepfake detection, Indian languages, Kathbath, IndicSynth, Opus compression, calibration, XLS-R, speaker-disjoint evaluation, cross-lingual transfer

I. INTRODUCTION

Neural voice cloning can imitate speakers from short enrollments and can be delivered as compressed chat audio. Major anti-spoofing benchmarks remain concentrated in English or Mandarin. VaaniQ studies Hindi, Marathi, and Tamil on a bounded speaker-disjoint benchmark and measures codec shift, cross-lingual transfer, and confidence calibration. It does not claim universal fake-voice detection. Telugu is out of project scope.

Contributions: (1) a versioned V1 publication corpus with paired Opus twins and held-out n=584; (2) complete RQ1–RQ4 results including negative calibration transfer and catastrophic English→Indic transfer retained honestly; (3) a reproducible research stack with frozen manifests and automated validation. Non-claims: not the first Indic detector; V1 source–label confounding is disclosed.

RQ1: Opus compression effects? RQ2: English-only versus multilingual transfer? RQ3: unseen-language (LOLO) generalisation? RQ4: validation-selected calibration under shift? RQ1–RQ4 are complete on the bounded benchmark.

II. RELATED WORK AND COMPARATIVE STUDY

A. Literature Axes

AASIST integrates spectro-temporal graph attention for anti-spoofing [1]. XLS-R provides cross-lingual self-supervised speech representations [2]. IndicSUPERB/Kathbath [3],[6] and IndicSynth [4],[7] supply Indic real and synthetic speech. Pascu et al. study calibrated SSL deepfake detection without Indian languages or an isolated WhatsApp-Opus cell [5]. FLEURS supports cross-lingual evaluation [8]. Opus is defined in RFC 6716 [9]. ASVspoof 2019 LA remains the English control resource [10]. Indic-CodecFake/SATYAM targets codec synthesis—a different attack surface than messaging-style Opus transport of cloning/TTS fraud audio—and does not couple Opus isolation with calibration analysis.

B. Feature Comparison Across Related Work

Cross-paper EER comparison is invalid under mismatched corpora. Table II compares evaluation axes only. VaaniQ does not claim to be first at Indic detection or calibration; the defensible contribution is their combination with isolated Opus and cross-lingual evaluation (proposal §5.7).

Axis summary: ASVspoof+AASIST supplies the non-Indic architecture benchmark; Pascu et al. supply calibration precedent without an isolated

Opus cell; IndicSynth supplies Indic synthetic data; SATYAM studies codec synthesis rather than Opus transport; VaaniQ combines hi/mr/ta, a 16 kbps Opus cell, calibration, and LOLO evaluation.

C. Same-Test Empirical Comparison (n=584)

All systems below share identical V1 held-out clips, labels, threshold 0.5, and score contract—the only fair numeric comparison in this paper. Matching 54.79% accuracy for LFCC-GMM and RawNet2-style at threshold 0.5 is a documented coincidence; EER and ROC-AUC differ materially. Faithful RawNet2 remains PENDING.

TABLE I. COMPARATIVE STUDY — SAME VAANIQ V1 TEST

Model	Acc / EER	AUC / min-DCF
LFCC-GMM	54.79% / 23.48%	0.8195 / 0.4962
RawNet2-style approx.	54.79% / 43.18%	0.5845 / 0.9621
Acoustic AASIST-comp.	91.61% / 6.56%	0.9729 / 0.7841
Frozen XLS-R + head	92.12% / 6.88%	0.9828 / 0.3144
English-only control	54.79% / 76.56%	0.162 / —

D. In-Domain versus Leave-One-Language-Out

TABLE II. COMPARATIVE VIEW — IN-DOMAIN VS LOLO

Language	In-domain Acc / EER	LOLO Acc / EER
Hindi	90.99% / 7.04%	78.83% / 21.83%
Marathi	93.29% / 6.25%	93.29% / 7.14%
Tamil	90.91% / 7.14%	93.94% / 6.35%

Hindi zero-shot collapse (90.99% in-domain → 78.83% LOLO) is the standout RQ3 finding and caution against treating Indian languages as a homogeneous transfer class.

III. DATASET

A. Bounded V1 Benchmark

The persisted source subset contains 1800 original clips (3.15 h): 900 Kathbath REAL and 900 IndicSynth FAKE, balanced across Hindi, Marathi, and Tamil. Paired validation/test 16 kbps libopus twins expand evaluation to 2,346 instances. Held-out test n=584. Bonafide clips follow Kathbath packaging terms; generated clips follow IndicSynth CC BY-NC 4.0 for non-commercial academic research.

V1 has a structural weakness: REAL equals Kathbath and FAKE equals IndicSynth. Source identity is therefore associated with class. Speaker-disjoint splitting protects against speaker leakage but does not remove source-domain shortcuts.

B. Partial V2 External-Source Pilot

Benchmark V2 is a partial external-source pilot. It adds an independent real-speech source but does not eliminate source-label confounding. Current cells include Kathbath×real, IndicSynth×fake, and FLEURS×real. Generator-disjoint evaluation has n=0 and remains PENDING.

TABLE III. DATASET COMPOSITION

Benchmark	Source / Class	Count / Status
V1	Kathbath / REAL	900 / COMPLETE
V1	IndicSynth / FAKE	900 / COMPLETE
V2	Kathbath / REAL	1,177 / PARTIAL
V2	IndicSynth / FAKE	1,169 / PARTIAL
V2	FLEURS / REAL	50 / PARTIAL

IV. METHODOLOGY

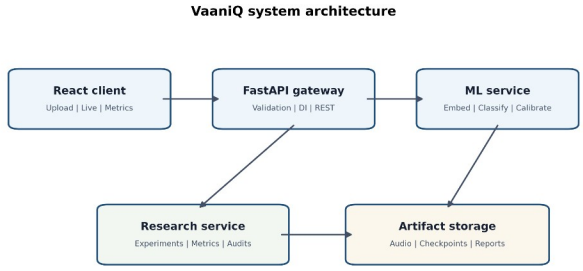


Fig. 1. VaaniQ architecture: preprocess → acoustic or frozen XLS-R → head → calibration.

A. Acoustic Baseline V1

Baseline V1 uses a deterministic 1024-D acoustic embedding and an AASIST-compatible NumPy classification head. It is not canonical AASIST and not XLS-R. Decision threshold 0.5 on the fake softmax score. Label 0=REAL, 1=FAKE; higher score_fake means more fake.

B. Frozen XLS-R Main

Frozen facebook/wav2vec2-xls-r-300m, last-layer mean pooling, cached features, and a lightweight trainable head. Backbone weights are not updated. Production calibration is global temperature scaling selected on validation.

C. Baselines and Controls

LFCC-GMM is a complete weak baseline. A RawNet2-style approximate baseline must not be called faithful RawNet2 (PENDING). An English-only ASVspoof LA control is evaluated on the same Indic held-out test.

D. Compression and Metrics

ffmpeg/libopus at 16 kbps provides a WhatsApp-style Opus simulation, not WhatsApp transport. EER, ROC-AUC, and normalized min-DCF share score direction. min-DCF uses P_target=0.05, C_miss=1, C_fa=1.

V. PRIMARY RESULTS

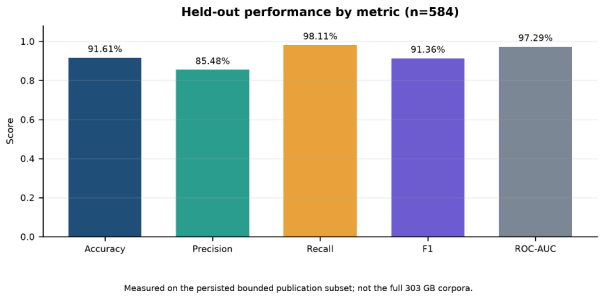


Fig. 2. Held-out performance overview.

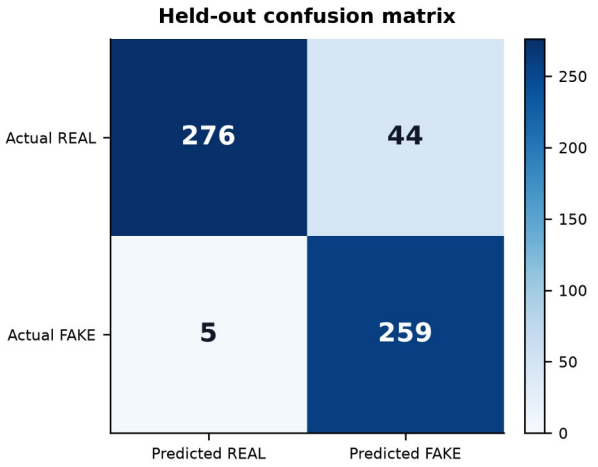


Fig. 3. Confusion matrix (Baseline V1).

TABLE IV. CANONICAL HELD-OUT RESULTS (THRESHOLD 0.5)

System	n	Acc / EER / AUC / min-DCF
Acoustic Baseline V1	584	91.61% / 6.56% / 0.9729 / 0.7841
Frozen XLS-R	584	92.12% / 6.88% / 0.9828 / 0.3144

Frozen XLS-R improves ranking performance while thresholded accuracy remains broadly comparable to Baseline V1.

VI. COMPRESSION (RQ1)

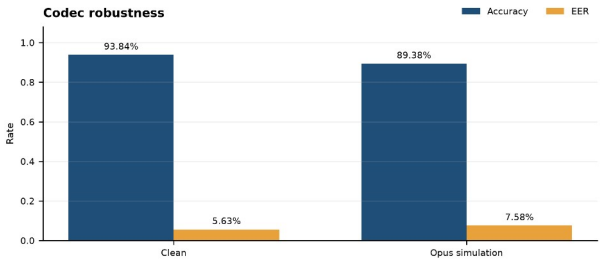


Fig. 4. Clean versus Opus condition breakdown.

TABLE V. RQ1 COMPRESSION (WHATSAPP-STYLE OPUS 16 KBPS)

Model	Clean Acc / EER	Opus Acc / EER
Acoustic	93.84% / 5.63%	89.38% / 7.58%
Frozen XLS-R	91.44% / 8.13%	92.81% / 6.06%

Compression effects are model-dependent on this benchmark.

VII. ENGLISH-ONLY TRANSFER (RQ2)

TABLE VI. RQ2 ENGLISH→INDIC TRANSFER

System	Acc / EER / AUC	Note
English-only control	54.79% / 76.56% / 0.162	All REAL at thr 0.5
Multilingual Baseline V1	91.61% / 6.56% / 0.9729	Same Indic test

English-only anti-spoofing fails to transfer. Official scores are not polarity-flipped to inflate AUC.

VIII. CROSS-LINGUAL TRANSFER (RQ3)

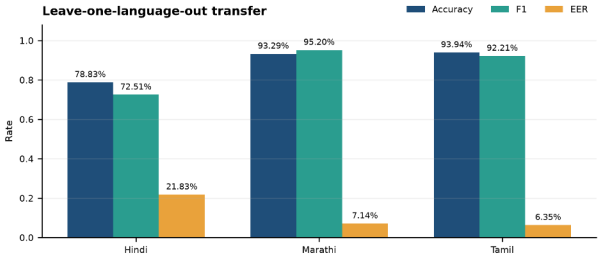


Fig. 5. Cross-lingual / language transfer view.

TABLE VII. RQ3 LEAVE-ONE-LANGUAGE-OUT (ROUND 3)

Held-out language	Accuracy	EER
Hindi	78.83%	21.83%
Marathi	93.29%	7.14%
Tamil	93.94%	6.35%

Transfer is strongly asymmetric; Hindi is the weakest zero-shot cell.

IX. CALIBRATION (RQ4)

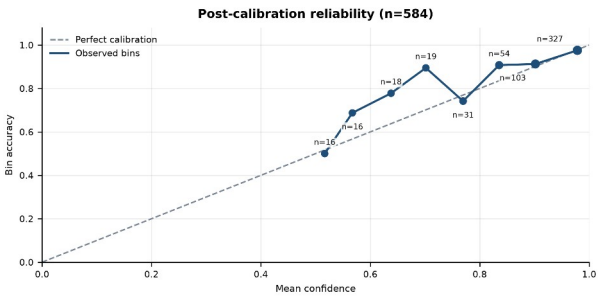


Fig. 6. Reliability diagram.

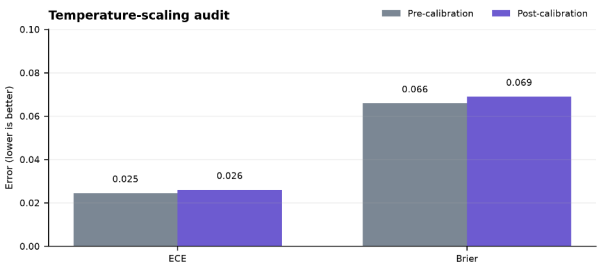


Fig. 7. Pre/post calibration comparison.

TABLE VIII. RQ4 CALIBRATION AUDIT

Quantity	Value	Role
Pre-cal test ECE	0.0245	Held-out uncalibrated
Post-cal test ECE	0.0260	After val-selected strategy
Selection rule	validation ECE only	No test peeking for production

Validation-selected fine-grained calibration slightly worsened held-out ECE (0.0245 → 0.0260). This negative transfer result is retained.

X. EXTERNAL VALIDITY AND V2 PILOT

TABLE IX. REMAINING GENERALIZATION STATUS

Eval	n	Status / Result
V2 full test	593	PARTIAL / not confound-free
FLEURS unseen-real	9	PILOT / 55.6% pipeline only
Generator-disjoint	0	PENDING / no result
V2 source probe	593	PARTIAL / 98.48% source identity
Faithful RawNet2	-	PENDING / not implemented

XI. DISCUSSION

Strong V1 discrimination coexists with structural threats. Catastrophic English→Indic transfer supports multilingual training for this domain. Asymmetric LOLO results caution against treating Indian languages as a homogeneous transfer class. Negative calibration transfer shows validation ECE gains require held-out audits. The paper therefore scopes its contribution to the reproducible machine-evaluation evidence reported here.

XII. LIMITATIONS

Key threats: (1) V1 source–label confound; (2) bounded size; (3) academic≠scam audio; (4) simulated≠real WhatsApp transport; (5) three-language scope; (6) non-canonical AASIST / pending faithful RawNet2; (7) incomplete V2; (8) source probe 98.48%; (9) FLEURS n=9; (10) generator-disjoint n=0; (11) calibration-transfer uncertainty; (12) unresolved val/test heterogeneity; (13) no universal unseen-generator claim. Strong EER can coexist with high min-DCF under a 5% target prior.

XIII. ETHICS

Audio and weights are excluded from version control. Kathbath access conditions and IndicSynth non-commercial restrictions are respected. Dual-use risks of detection systems are acknowledged; release emphasizes research evaluation.

XIV. REPRODUCIBILITY

Authoritative metrics are frozen in artifacts/final_results_manifest.json at commit 084bd47ca6ca1b69a7cdbf424e2946f3794c2a95. Integrity verification: scripts/verify_research_integrity.py. Documents must consume the frozen manifest rather than superseded checkpoints.

XV. FUTURE WORK

Expand confound-controlled and generator-disjoint evaluation; implement faithful RawNet2 and canonical AASIST; study real messaging-channel audio beyond Opus simulation.

XVI. CONCLUSION

VaaniQ establishes a reproducible multilingual framework for studying audio-deepfake detection under language, codec, and confidence-calibration shift. Acoustic and frozen XLS-R pipelines achieve strong held-out discrimination on bounded V1, while transfer, calibration, and source-domain experiments expose important limits. Evidence supports multilingual machine evaluation but does not establish universal source- or generator-independent detection.

REFERENCES

[1] J. Jung et al., AASIST: Audio anti-spoofing using integrated spectro-temporal graph attention networks, in Proc. ICASSP, 2022.
[2] A. Babu et al., XLS-R: Self-supervised cross-lingual speech representation learning at scale, arXiv:2111.09296, 2021.
[3] T. Javed et al., IndicSUPERB: A speech processing universal performance benchmark for Indian languages, arXiv:2208.11761, 2022.
[4] D. V. Sharma, V. Ekbote, and A. Gupta, IndicSynth: A large-scale multilingual synthetic speech dataset, ACL, 2025.
[5] O. Pascu et al., Towards calibrated and explainable audio deepfake detection, in Proc. Interspeech, 2024.
[6] AI4Bharat, Kathbath dataset, Hugging Face: ai4bharat/Kathbath.
[7] D. V. Sharma et al., IndicSynth dataset, Hugging Face: vdivyasharma/IndicSynth.
[8] A. Conneau et al., FLEURS: Few-shot learning evaluation of universal representations of speech, arXiv:2205.12446, 2022.
[9] IETF RFC 6716, Definition of the Opus Audio Codec, 2012.
[10] ASVspoof 2019, logical access evaluation plan; English-only control uses a streaming subset of Bisher/ASVspoof_2019_LA (ODC-By).